Sample questions for CO1 – Assessment Tool 3 – Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

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CO1 - Assessment Tool-3

① <u>cloud Computing</u> :- The way of Providing Cloud services like Servers, Power, GPUs, Network etc through Over an internet is said to be cloud computing.

<u>Characteristics of cloud computing</u>

* <u>As-Pay-as-you</u> Use : The User Can be used the required amount of web service where Can be Paid for how much they use.
* <u>Elasticity</u> :- The Servers Can be maintained as scalable up and down as user required at busy times.
* <u>Access</u> :- The user Can be Accessible to server from Anywhere across the world through Authentication.

eg:- user → india;
server → USA

* <u>cloud Collaboration</u> :- As Cloud user can Provide extra Services, but instead of buying new server Can Collab with other Cloud Partner to improve scalability.

→ <u>Cloud</u> :- Can be Accessible, unlimited, Scalable, easier, less Cost.

→ <u>On-Premise</u> :- Limited, less scalable, Person-based acess.

(2) Evolution of Cloud Computing

* Early time Cloud :- [1980's]

The thing is where the Computer is Centralized and used all together which is very large to Carry from one place to other.

* Client & server : [1990's]

* The Applications work on Computer is managed in a Computer between client and server.

* The size of hardware is reduced and RAMs are evolved, accessible by clients.

* Software Evolution [2000's]

* The softwares are Created which is accessible over internet as installation is just to use as Company made.

* Server's Improvement [2010's]

- Servers are made which are in data Centere own by a Company that Provides access to all user through feature like backup, storage, acess from anywhere.

* Improved scalability & Reliable

* Fast access, manage their data.
* Reliable even many users access on same-time.
* Cloud can be rented with Person interaction.

③ Server Virtualization :- The Process of creating multiple virtualization on hardware (single physical hardware) is meant by server virtualization.

* improves Fast access, model services.
* Virtualization helps
  → Hardware utilizes more.
  → less Power optimization.

Virtualization	Cloud Computing
* Creates multiple virtuals on a single Hardware.	* cloud can be accessible over internet, not created
* More Power Consuming	* less Power, Cost Consuming
* It helps in working more storage, Network usage.	* Cloud takes efficient in using storage, Network.
* This works on using Hardware storage System.	* Cloud also can Provide own storage system.
eg:- Hard disk, CPUs	eg:- AWS, Oracle, Azure, etc.

④ **Data Centers** :- The main Central of cloud Providers is installed at one Place where they Contain servers, Data, Power, Network, GPUs, storage systems, etc is maintained.

Role of Delivery :-

* Now-a-days cloud is made as As-You-Pay-Us use meant the Amount U use can be Payable to cloud Provider.
* Where data centers has Centralized Cooling systems which are using billion litres of water per day to cool them.
* These are Available through over-internet without Person interaction.

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⑤ The Cloud is classified into Four main services.

① **Iaas** : Infrastructure as a Service
* Cloud can Provide infrastructure without user creation. eg :- OS, Virtualization.

② **Paas** :- Platform as a Service
* Provide a Platforms where all users can access without rent. eg:- Google search Engine, Browser.

③ **Saas** :- Software as a Service
* Provide Softwares for free, eg:- Gmail, 365 office. limited, Authorization.

④ **Xaas** :- Anything as a Service
* All Requirements like CPU, GPU, storage, Network, Power, etc.

eg:- Azure, AWS